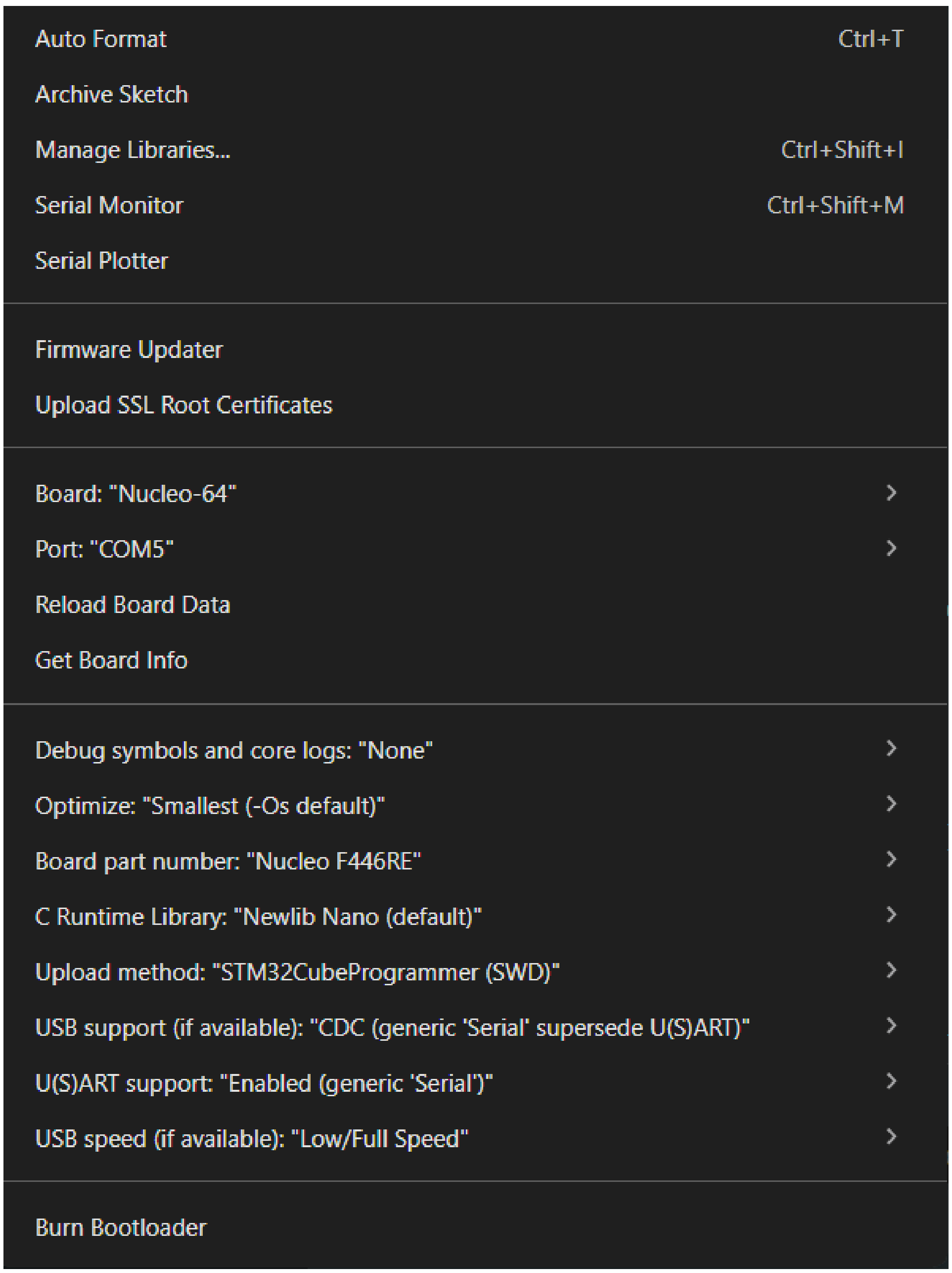
Smart Room Panel

STM32 + ESP32 + Firebase

DHT11 (temp/humidity) + PIR (motion) + Relay (light) + I2C LCD on an STM32 Nucleo F446RE, bridged to Firebase Realtime Database through an ESP32, with a web dashboard to view sensor data and toggle the light.

```
Web Dashboard  <-->  Firebase RTDB  <-->  ESP32 (WiFi)  <-- UART -->  STM32 F446RE
                                                                        |- DHT11
                                                                        |- PIR
                                                                        |- Relay
(light)
                                                                        |- I2C LCD
```



WiFi.h

Firebase_ESP_Client.h

SoftwareSerial.h

DHT11.h

Wire.h

LiquidCrystal_I2C.h

Files

- stm32_sketch/stm32_sketch.ino — upload to the Nucleo F446RE
- esp32_sketch/esp32_sketch.ino — upload to the ESP32
- web_dashboard/index.html — open in a browser, or host on Firebase Hosting / GitHub Pages

1. Wiring

DHT11 → STM32

DHT11	STM32
VCC	5V
GND	GND
DATA	PB0

PIR sensor → STM32

PIR	STM32
VCC	5V
GND	GND
OUT	PB1

Relay module (light) → STM32

Relay	STM32
VCC	5V
GND	GND

Relay	STM32
IN	PA5

⚠️ **Never wire mains voltage directly to the STM32.** Use a relay module rated for your light's voltage/current, and keep the mains side fully isolated.

16x2 I2C LCD → STM32

LCD	STM32
VCC	5V
GND	GND
SDA	PB9
SCL	PB8

ESP32 ↔ STM32 (UART bridge)

ESP32	STM32
GND	GND
`TX2 (GPIO17)`	`PA10 (RX)`
`RX2 (GPIO16)`	`PA9 (TX)`

2. Arduino IDE setup

STM32 side

Board: Nucleo-64, Board part number: Nucleo F446RE, Upload method: STLink

Libraries to install (Library Manager):

- DHT sensor library (Adafruit) + Adafruit Unified Sensor
- LiquidCrystal I2C (Frank de Brabander)
- SoftwareSerial (STM32duino-compatible version)

ESP32 side

Boards Manager URL: `https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json`

Board: ESP32 Dev Module

Library to install: Firebase ESP Client (by mobizt)

3. Firebase project setup

- Go to Firebase Console (https://console.firebase.google.com) → Create project
- Realtime Database → Create database → start in test mode (tighten rules later)

- Authentication → Sign-in method → enable Email/Password, and enable Anonymous too (the web dashboard uses anonymous sign-in by default)
- Authentication → Users → add one user (email/password) — this is what the ESP32 sketch uses
- Project settings → General → copy the Web API Key and the Database URL
- Fill these into both esp32_sketch.ino and web_dashboard/index.html:
    - API_KEY / apiKey
    - DATABASE_URL / databaseURL
    - USER_EMAIL / USER_PASSWORD (ESP32 only)

4. Running it

- Upload stm32_sketch.ino to the Nucleo (Serial Monitor at 9600 baud shows debug logs)
- Upload esp32_sketch.ino to the ESP32 (Serial Monitor at 115200 baud — wait for the IP address and "Firebase initialized")
- Wire the ESP32 and STM32 together as above
- Open web_dashboard/index.html in a browser (double-click works, no server needed)
- You should see live temperature/humidity/motion, and the rocker switch should toggle your light